Second-order interpolation conditions for univariate functions, and application to optim[redacted]

Benelux Meeting 2024

This nice guy!

https://nizarbousselmi.github.io/

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Overview

1. On the importance of interpolation conditions in optimization
   →E.g.: Performance of Gradient method on smooth convex functions

2. Second order conditions for univariate functions
   →The « tricky part » of the presentation

3. Some applications
   → Finally, some optimization!

Worst-case performance analysis of algorithms

How to obtain guarantees on the worst-case performance of optimization methods on a given function class?

Input:

- Method → relationship between iterates $x_0, x_1, \cdots$ $\quad x_{k+1} = x_k - hg_k$
- Class → inequalities satisfied by iterates, gradients, function values $\quad f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$

Combination of the inequalities

Theorem 2.1.13 *If $f \in \mathcal{F}_L^{1,1}(R^n)$ and $0 < h < \frac{2}{L}$ then the gradient method generates the sequence $\{x_k\}$ such that*

$$f(x_k) - f^* \leq \frac{2(f(x_0) - f^*) \parallel x_0 - x^* \parallel^2}{2 \parallel x_0 - x^* \parallel^2 + (f(x_0) - f^*)h(2 - Lh)k}.$$

Proof:
Denote $r_k = \parallel x_k - x^* \parallel$. Then

$$\begin{aligned} r_{k+1}^2 &= \parallel x_k - x^* - hf'(x_k) \parallel^2 \\ &= r_k^2 - 2h\langle f'(x_k), x_k - x^* \rangle + h^2 \parallel f'(x_k) \parallel^2 \\ &\leq r_k^2 - h(\tfrac{2}{L} - h) \parallel f'(x_k) \parallel^2 \end{aligned}$$

(we use (2.1.8) and $f'(x^*) = 0$). Therefore $r_k \leq r_0$. In view of (2.1.6) we have:

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) + \langle f'(x_k), x_{k+1} - x_k \rangle + \tfrac{L}{2} \parallel x_{k+1} - x_k \parallel^2 \\ &= f(x_k) - \omega \parallel f'(x_k) \parallel^2, \end{aligned}$$

$$(\ldots)$$

Output: A priori guarantees on the quality of $f(x_N) - f^*, ||x_N - x^*||, \cdots$ after N iterations.

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Tight *discrete* representation of the function classes

Combination of the inequalities

Sources of conservatism:
- Non tight inequalities
- Unoptimal combination of the inequalities

Output: A priori guarantees on the quality of $f(x_N) - f^*, ||x_N - x^*||,$ after N iterations.

Tight *discrete* representation of function classes

The constraint characterizing a class can only be imposed on a finite set of data $S = \{(x_i, g_i, f_i\}_{i \in [N]}$

Example: the class of L-smooth convex functions

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$$
$$||\nabla f(x) - \nabla f(y)|| \leq L||x - y||$$

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{||\nabla f(x) - \nabla f(y)||^2}{2L}$$

Equivalent constraints

Necessary constraint

Necessary constraint

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle$$
$$||g_i - g_j|| \leq L||x_i -$$

$$\geq f_j + \langle g_j, x_i - x_j \rangle + \frac{||g_i - g_j||^2}{2L}$$

Non equivalent constraints

Some constraints are stronger than others

Necessary and sufficient for interpolation by a function of interest

Which discretized version of a characterization is an interpolation constraint?

$$\boxed{\begin{array}{l} f_i \geq f_j + \langle g_j, x_i - x_j \rangle \\ ||g_i - g_j|| \leq L||x_i - x_j|| \end{array}}$$

$1 \leq g_2 \leq 2$

$f_2 \geq 1$

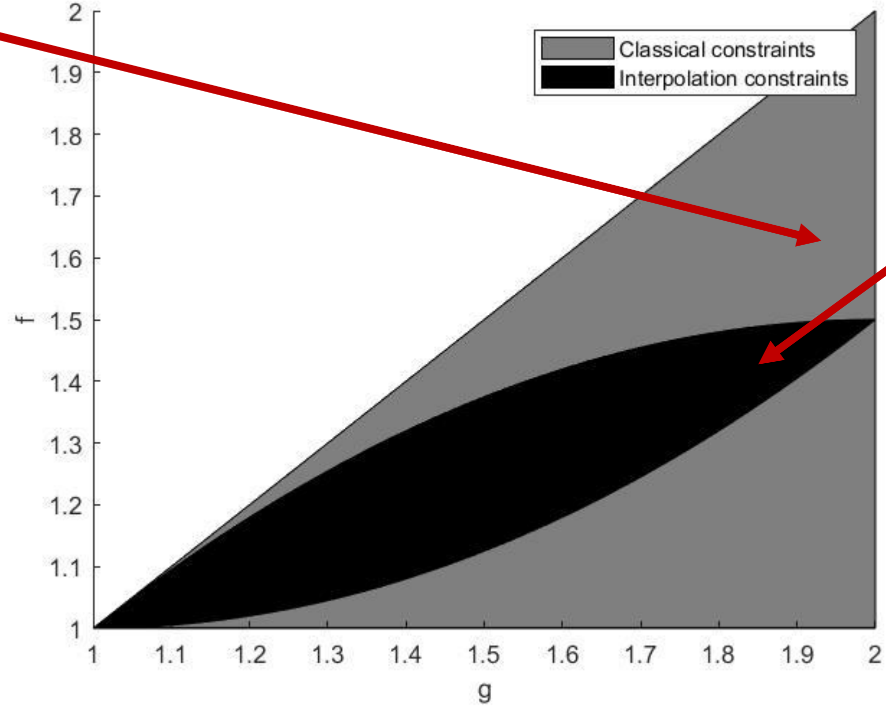
$f_2 \leq g_2 \leq 2$

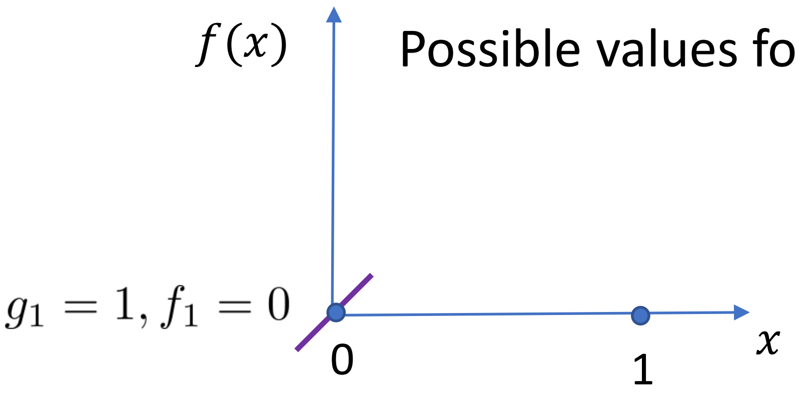
$$\boxed{f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \frac{||g_i - g_j||^2}{2L}}$$

$1 \leq g_2 \leq f_2$

$f_2 \geq 1 + (g_2 - 1)^2/2 \geq 1$

$f_2 \leq g_2 - (g_2 - 1)^2/2 \leq 1.5$





Possible values for f(1),g(1)?

Worst-case performance analysis of algorithms

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Interpolation constraints: Tight *discrete* representation of the function classes

Combinaison of the inequalities

PEP framework: Automatic computation of guarantees by an optimal combinaison of the inequalities [THG17]

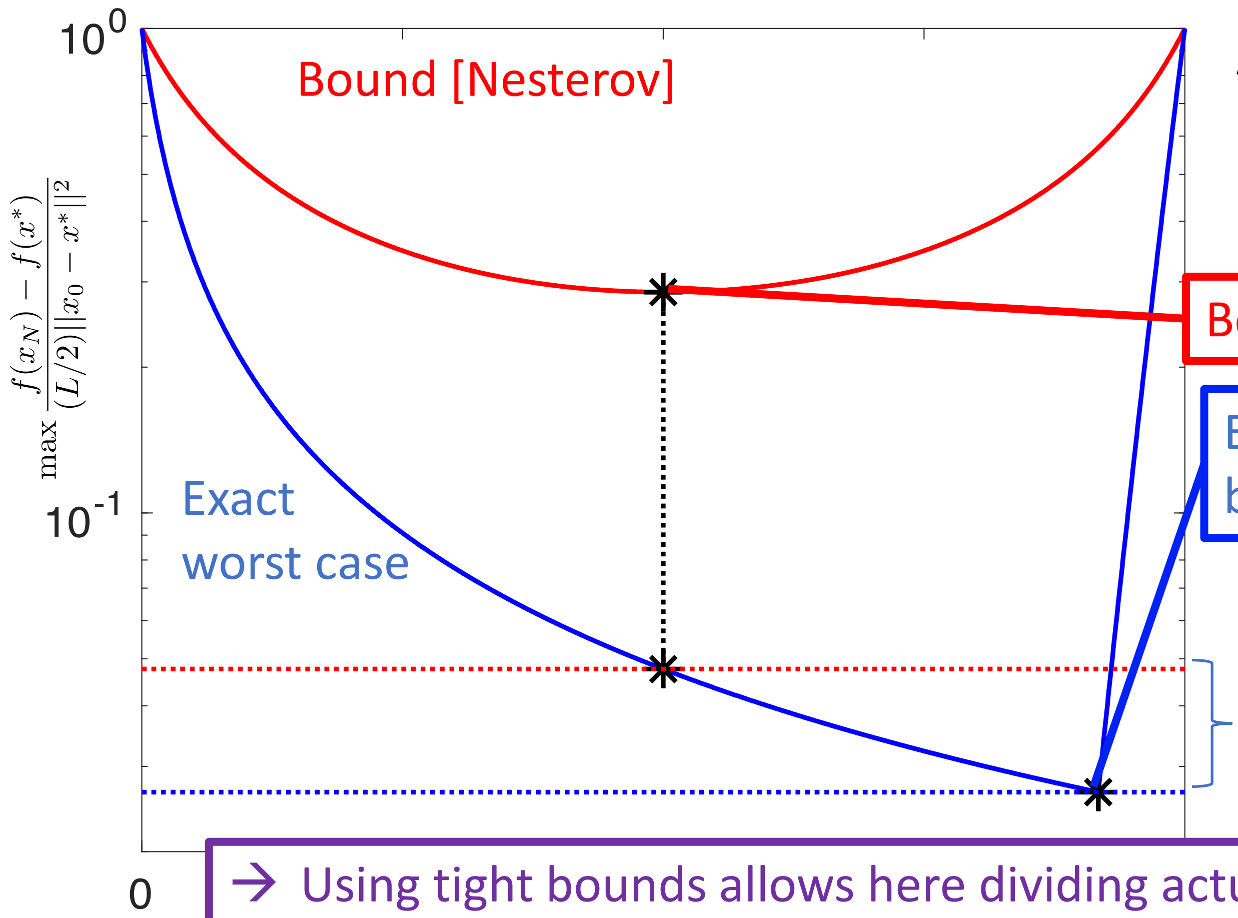
Sources of conservatism:
- Non tight inequalities
- Unoptimal combination of the inequalities

Output: A priori **tight** guarantees on the quality of $f(x_N) - f^*, ||x_N - x^*||,$ after N iterations.

[THG17] A. B. Taylor, J. M. Hendrickx, and F. Glineur, Exact worst-case performance of first-order methods for composite convex optimization, 2017

Why are tight bounds important?

N = 10 iterations



Accuracy guarantee

Bound-optimized step

Exact worst-case based Optimal step

Performance gap $\sim \mathbf{2}$

→ Using tight bounds allows here dividing actual (worst) computation cost by $\sim \mathbf{2}$ by optimal tuning

State of the art… order 2 is missing!

- ORDER 0 : Lipschitz functions,… → OK ☺

- ORDER 1 : Smooth functions

  (Strongly/weakly) convex functions

  Indicators function → OK ☺

  Bounded domains/subdifferential

  Linear operators

⇒ First-order methods: GD, FGD, proximal gradient,… → OK ☺

- ORDER 2: Functions with Lipschitz hessian → NOT OK

$$||\nabla^2 f(x) - \nabla^2 f(y)|| \leq M||x-y||$$

⇒ Newton's method, Damped NM
Cubic NM→ NOT OK

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Second order interpolation conditions for univariate functions

Global characterization of $\mathcal{H}_M$ $(\forall x, y)$

$f : \mathbb{R} \to \mathbb{R}$

$$|f''(x) - f''(y)| \leq M|x-y|$$

A widely used characterization

$$|f(x) - f(y) - g(y)(x-y) - \frac{1}{2}h(y)(x-y)^2| \leq \frac{M}{6}|x-y|^3$$

Interpolation conditions $(S = \{(x_i, g_i, h_i, f_i)\}_{i\in[N]})$

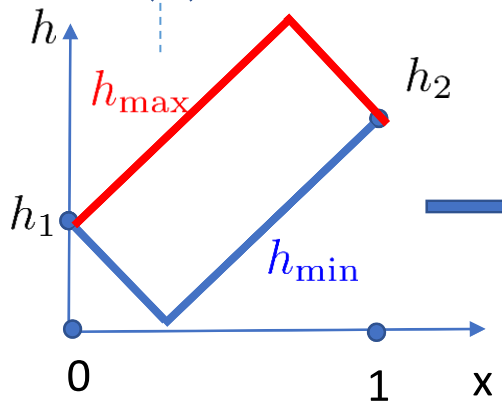
Discretization?

$$|h_i - h_j| \leq M|x_i - x_j|$$

$$|f_i - f_j - g_j(x_i - x_j) - \tfrac{1}{2}h_j(x_i - x_j)^2| \leq \tfrac{M}{6}|x_i - x_j|^3$$

New idea for 1D function:

1. « Extremal » hessians/gradients associated to a function in $\mathcal{H}_M$ consistent with S.
2. Integration of these expressions



$$\int_0^1 h_{\min}(x)dx \leq g_2 - g_1 \leq \int_0^1 h_{\max}(x)dx$$

Second order interpolation conditions for univariate functions

Global characterization of $\mathcal{H}_M$ $(\forall x, y)$

$f : \mathbb{R} \to \mathbb{R}$

$$|f''(x) - f''(y)| \le M|x-y|$$

$$|f(x) - f(y) - g(y)(x-y) - \frac{1}{2}h(y)(x-y)^2| \le \frac{M}{6}|x-y|^3$$

Interpolation conditions $(S = \{(x_i, g_i, h_i, f_i)\}_{i\in[N]})$

Theorem 1. *A set* $S = \{(x_i, g_i, h_i, f_i)\}_{i\in[N]}$ *is* $\mathcal{H}_M$*-interpolable, if, and only if,* $\forall i,j \in [N]$

$$|\Delta h_{ij}| \le M|\Delta x_{ij}|$$

$$T^f_{ij} \ge -\frac{M}{6}|\Delta x_{ij}|^3 + \frac{\left(T^g_{ij} + \frac{M}{2}\Delta x_{ij}|\Delta x_{ij}|\right)^2}{2\left(\Delta h_{ij} + M|\Delta x_{ij}|\right)} + \frac{\left(\Delta h_{ij} + M|\Delta x_{ij}|\right)^3}{96M^2},$$

where

$$\Delta x_{ij} = x_j - x_i$$
$$T^f_{ij} = f_j - f_i - g_i\Delta x_{ij} - \frac{h_i}{2}\Delta x^2_{ij}$$
$$T^g_{ij} = g_j - g_i - h_i\Delta x_{ij}$$
$$\Delta h_{ij} = h_j - h_i.$$

Positive terms: strengthen the constraint

Tight discrete representation of $\mathcal{H}_M$: done!
→ How to exploit it?

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Tackling Newton's method

From $|T_{ij}^f| \leq \frac{M}{6}|\Delta x_{ij}|^3$ to $T_{ij}^f \geq -\frac{M}{6}|\Delta x_{ij}|^3 + \frac{\left(T_{ij}^g + \frac{M}{2}\Delta x_{ij}|\Delta x_{ij}|\right)^2}{2(\Delta h_{ij} + M|\Delta x_{ij}|)} + \frac{(\Delta h_{ij} + M|\Delta x_{ij}|)^3}{96M^2}$

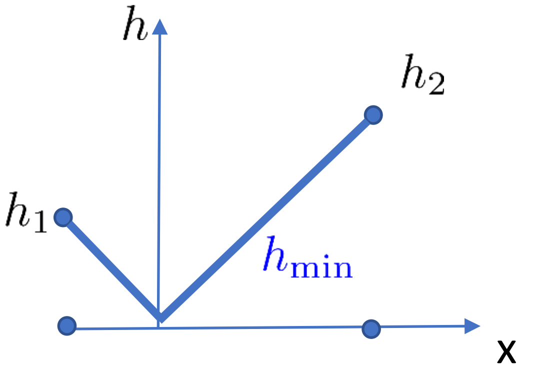
In combination with the PEP framework→ exact results!
(In 1D, lower bound for all dimensions)
If matches known upper bound→ exact multivariate bound)!

Newton's Method: one iteration:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

PEP with interpolation conditions
→ matching lower bound!
→ Identification of worst-case function:

$$f_1(x) = \begin{cases} \frac{M}{6}x^3 + \frac{\mu}{2}x^2 & \text{if } x \leq 0 \\ -\frac{M}{6}x^3 + \frac{\mu}{2}x^2 & \text{if } x \geq 0. \end{cases}$$



Known upper bound [Nes18]:

$$||x_{k+1} - x^*|| \leq \frac{\frac{M}{\mu}||x_k - x^*||^2}{2\left(1 - \frac{M}{\mu}||x_k - x^*||\right)} := R_k$$

This bound is tight for all dimensions!

[Nes18] Y. Nesterov et al., Lectures on convex optimization. Vol. 137. Springer, 2018.

Tackling Newton’s method

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Other criteria, **derived analytically :**

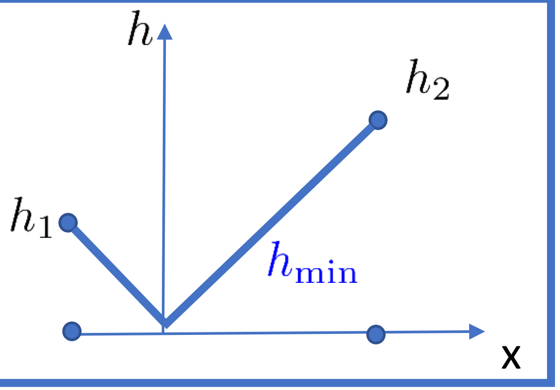
$$f(x_{k+1}) - f(x^*) \leq \frac{M}{6}R_k^3 + \frac{\mu}{2}R_k^2,$$

$$||\nabla f(x_{k+1})|| \leq \frac{M}{2}R_k^2 + \mu R_k,$$

$$\lambda_{\min}(\nabla^2 f(x_{k+1})) \geq -MR_k + \mu.$$

PEP with interpolation conditions
→ matching lower bound!
→ **Same worst-case function:**

$$f_1(x) = \begin{cases} \frac{M}{6}x^3 + \frac{\mu}{2}x^2 & \text{if } x \leq 0 \\ -\frac{M}{6}x^3 + \frac{\mu}{2}x^2 & \text{if } x \geq 0. \end{cases}$$



These bounds are tight for all dimensions (and new)!

Some numerical results

From $|T_{ij}^f| \leq \frac{M}{6}|\Delta x_{ij}|^3$ to $T_{ij}^f \geq -\frac{M}{6}|\Delta x_{ij}|^3 + \frac{\left(T_{ij}^g + \frac{M}{2}\Delta x_{ij}|\Delta x_{ij}|\right)^2}{2(\Delta h_{ij} + M|\Delta x_{ij}|)} + \frac{(\Delta h_{ij} + M|\Delta x_{ij}|)^3}{96M^2}$

In combination with the PEP framework→ exact results!
(In 1D, lower bound for all dimensions)

Newton's damped Method: one iteration:
$$x_{k+1} = x_k - \alpha \frac{f'(x_k)}{f''(x_k)}$$

→ Exact worst-case in 1D
→ Identification of the best coefficient
→ Identification of the worst-case function.
→ Conjecture in the multivariate case

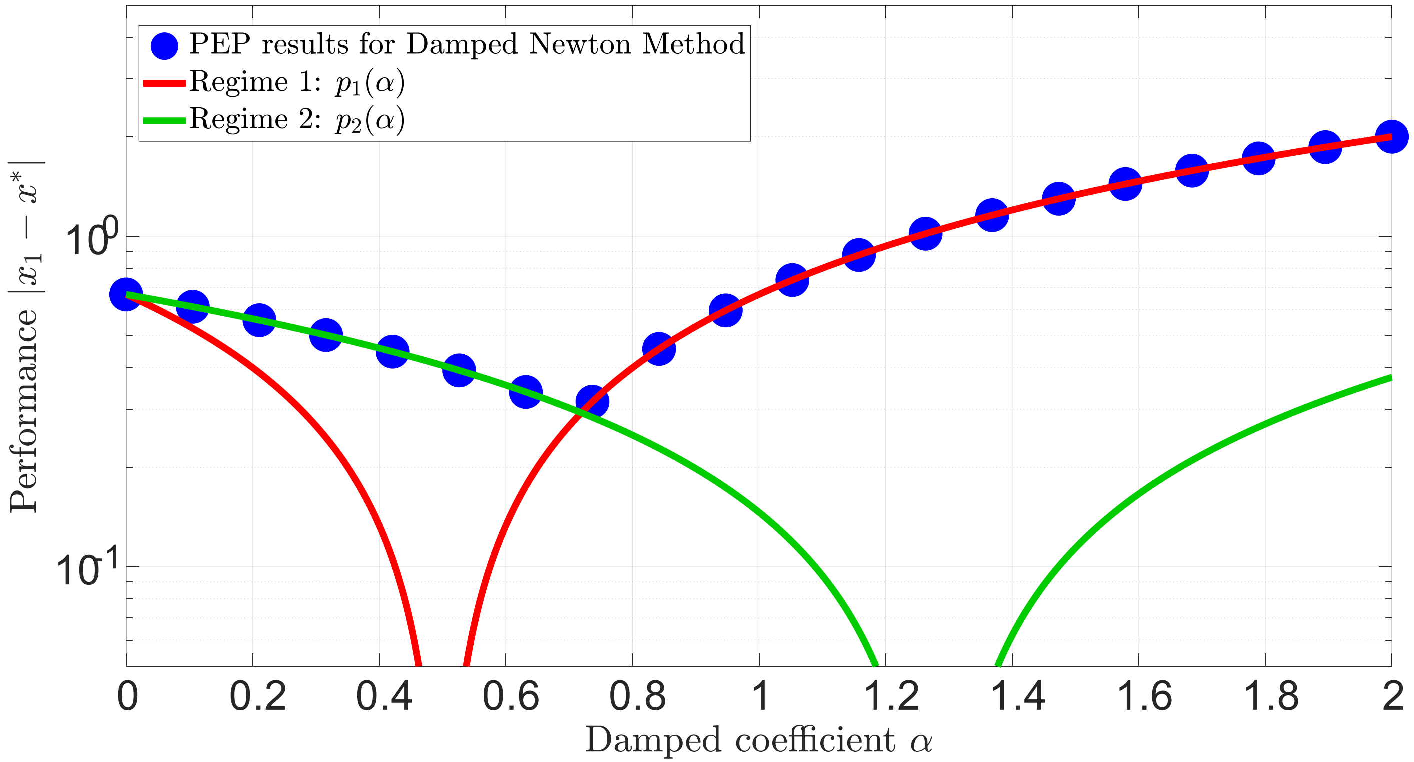


Fig. 1: $M = \mu = 1$ and $|x_0 - x^*| = \frac{2}{3}$

Some analytical results

From $|T_{ij}^f| \leq \frac{M}{6}|\Delta x_{ij}|^3$ to $T_{ij}^f \geq -\frac{M}{6}|\Delta x_{ij}|^3 + \frac{\left(T_{ij}^g + \frac{M}{2}\Delta x_{ij}|\Delta x_{ij}|\right)^2}{2(\Delta h_{ij} + M|\Delta x_{ij}|)} + \frac{(\Delta h_{ij} + M|\Delta x_{ij}|)^3}{96M^2}$

Can be used directly (by hand) to improve convergence proofs

Cubic Newton's Method: one iteration:

$$x_j = \arg\min_y \left(f_i + g_i(y - x_i) + \frac{1}{2}(y - x_i)^2 + \frac{M}{6}|y - x_i|^3\right)$$

Descent Lemma (decrease at each iteration) [Nes18]

$$f_j \leq f_i - \frac{M}{12}|\Delta x_{ij}|^3$$

Improved Descent Lemma:

$$f_j \leq f_i - \frac{M}{12}|\Delta x_{ij}|^3 - \frac{g_j^2}{2(M|\Delta x_{ij}| - \Delta h_{ij})} - \frac{1}{96M^2}\left(M|\Delta x_{ij}| - \Delta h_{ij}\right)^3$$

Positive terms!

[Nes18] Y. Nesterov et al., Lectures on convex optimization. Vol. 137. Springer, 2018.

Conclusion

One dimensional method→ Second-order interpolation conditions!

- Exact bounds in 1D
- Lower bounds for the multivariate case. If matching known upper bounds, exact multidimensional bounds!
- Improvement of analytical results

$$\boxed{T_{ij}^{f} \geq -\frac{M}{6}|\Delta x_{ij}|^3 + \frac{\left(T_{ij}^{g} + \frac{M}{2}\Delta x_{ij}|\Delta x_{ij}|\right)^2}{2(\Delta h_{ij} + M|\Delta x_{ij}|)} + \frac{(\Delta h_{ij} + M|\Delta x_{ij}|)^3}{96M^2}}$$

Future work and open questions:

- Extension to multidimensional functions
- Extension to higher orders
- A quest for the « right » function classes: second-order function classes with simpler discrete representation and interesting properties.

Thank you for your attention!

Any question?

References

[THG17] A. B. Taylor, J. M. Hendrickx, and F. Glineur, "Exact worst-case performance of first-order methods for composite convex optimization," SIAM J. on Optimization, 27(3), 2017.

[N18] Y. Nesterov et al., Lectures on convex optimization. Vol. 137. Springer, 2018.

The idea behind PEP

Worst-case instances are themselves solutions to optimization problems

Optimal combinaison of the inequalities

$$\max_{x_0,\cdots,x_N,x^*,f} f_N - f^*$$

$$\text{s.t. } f \in \mathcal{F}_L^{1,1}$$

$$x_{k+1} = x_k - \frac{g_k}{L}, k = 0, \cdots, N-1$$

$$||x^* - x_0||^2 \leq R^2$$

$$\nabla f(x^*) = 0$$

Tight description

Definition of the class (everywhere)

Definition of the method

How to ensure the tractability of a PEP?

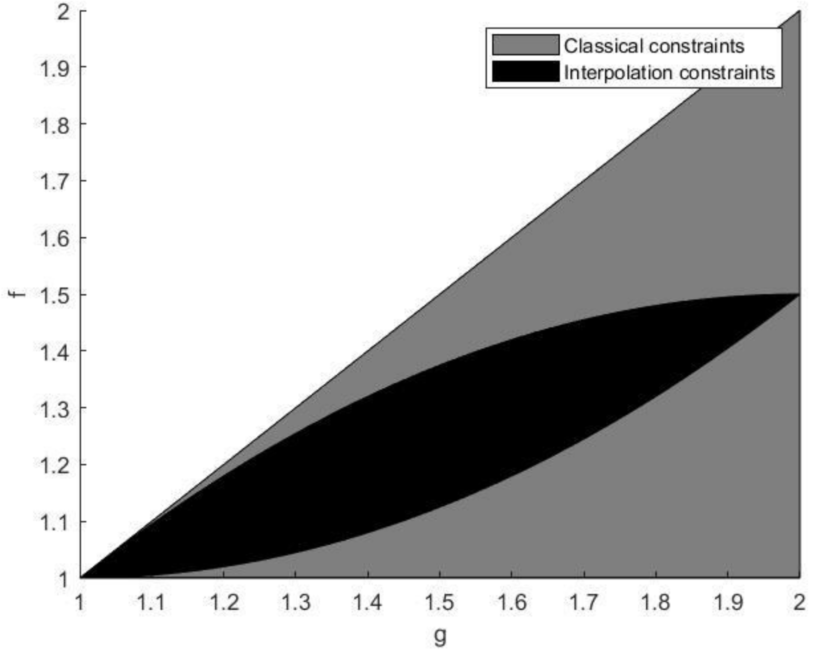
$$\max_{x_0,\cdots,x_N,x^*,f} f_N - f^*$$

$$\text{s.t. } f \in \mathcal{F}_L^{1,1}$$

Definition of the class

$$, k = 0, \cdots, N-1$$

Definition of the method



Bound obtained might be over conservative if the solution belongs to the grey area

Interpolation constraints to impose on a finite data set

Need for a tight discretized representation of the class

Why are tight bound important?

$$\max_{x_0,\cdots,x_N,x^*,f} f_N - f^*$$
$$\text{s.t. } f \in \mathcal{F}_L^{1,1}$$
$$x_{k+1} = x_k - \frac{g_k}{L}, k = 0, \cdots, N-1$$
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$$\nabla f(x^*) = 0$$

$$\max_{S=\{(x_i,f_i,g_i)\}_{i=0,\cdots,N,*}} f_N - f^*$$
$$\text{s.t. } f_i \geq f_j + \langle g_j, x_i - x_j\rangle + \frac{||g_i - g_j||^2}{2L}, \; i,j \in S$$
$$x_{k+1} = x_k - \frac{g_k}{L}, k = 0, \cdots, N-1$$
$$||x^* - x_0||^2 \leq R^2$$
$$g^* = 0$$

Linear in function values and Gram Matrix G (scalar products of iterates and subgradients)

Gram-representable performance measure, interpolation constraint, method

→ Exact performance guarantee by solving a semi-definite program

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$$\max_{S=\{(x_i,f_i,g_i)\}_{i=0,\cdots,N,*}} f_N - f^*$$

$$\text{s.t. } f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \frac{||g_i - g_j||^2}{2L}, \; i,j \in S$$

$$x_{k+1} = x_k - \frac{g_k}{L}, k = 0, \cdots, N-1$$

$$||x^* - x_0||^2 \leq R^2$$

$$g^* = 0$$

Linear in function values and Gram Matrix G

$$P = [x_0 \; g_0 \cdots g_N \; g_0^y \cdots g_N^y], \;\; G = P^T P$$

$$\max_{F,G} c^T F$$

$$\text{s.t. } b_{ij}^T F \geq Tr(A_{ij} G), \; \forall i,j$$

$$Tr(C_{ij} G) \leq R^2$$

Gram-representable performance measure, interpolation constraint, method

→ Exact performance guarantee by solving a semi-definite program